

Task B: Solution Scoping — GST Reconciliation Tool

Candidate Showcase Artifact | 20-Minute Interview Presentation **Topic:** How I Scoped, Designed, and Architected an AI-Assisted Compliance Solution **Project:** GST Reconciliation Tool & Compliance Auditor

1. Problem Statement

Under India's Goods and Services Tax (GST) regime, businesses can only claim **Input Tax Credit (ITC)** on vendor purchases if their suppliers have properly filed those invoices on the government GST portal (**GSTR-2B**).

In practice, company internal accounting books (**Purchase Registers**) and government portal files (**GSTR-2B**) diverge significantly due to:

- Diverse ERP invoice numbering schemes (e.g., INV/23-24/0043 vs. 43)
- Vendor legal name variations (e.g., Tata Consultancy Services Ltd vs. TATA CONSULTANCY SERVICES PRIVATE LIMITED)
- Tax rate discrepancies (e.g., 12% vs. 18%)
- Amount rounding and data entry errors
- Unfiled supplier invoices (MISSING_IN_2B)

If an organization claims credit on unfiled or mismatched invoices, **Section 16(2)(aa) of the CGST Act** mandates repayment of the tax plus an **18% per annum interest penalty**, with potential audit penalties up to 100%.

2. User & Customer Need

Target Users

- 1. Corporate Finance Controllers & CFOs:** Need high-level visibility into total financial exposure, supplier compliance risk, and potential working capital blockage.
- 2. Tax Managers & Chartered Accountants (CAs):** Need automated monthly reconciliation that replaces 40–80 hours of manual Excel VLOOKUPS with instant, verifiable audit trails.
- 3. Accounts Payable (AP) Teams:** Need a prioritized list of non-compliant suppliers to withhold payment or issue debit notes before tax filing deadlines.

Core User Needs

- **Multi-Format Ingestion:** Accept raw exports from ERPs (SAP, Tally, Zoho) and the GST portal without requiring pre-formatting into a rigid template.
- **Deterministic Mathematical Precision:** Financial totals and tax penalties must be calculated with 100% precision. Probabilistic LLM arithmetic is unacceptable in accounting.
- **Explainable Discrepancy Classification:** Rather than a binary "match/no match", users need clear categorization (e.g., rate mismatch vs. timing difference) with a plain-language audit reason.
- **Human Authority:** Tax teams must retain final sign-off authority on ambiguous invoices before credits are permanently disallowed or claimed.

3. Goals

Goal	Success Metric
High Reconciliation Accuracy	Achieve > 90% accuracy on status matching and discrepancy categorization across real-world edge cases.
Zero Financial Hallucination	0.00% arithmetic variance on calculated tax liabilities. All math executed via Python Decimal.
High-Throughput Scalability	Process up to 1,000 invoices in under 5.0 seconds on standard compute.
Cost & Token Efficiency	Minimizing LLM token usage by resolving > 75% of clean records through deterministic rules first.
Zero-Crash Resilience	The system must never crash if an external API key is missing or internet connectivity drops.

4. Requirements

Functional Requirements

- **FR-1 Multi-Format File Ingestion:** Ingest Purchase Register and GSTR-2B datasets in either .csv or multi-page .pdf formats.

- **FR-2 Data Normalization:** Standardize column headers, strip corporate legal suffixes (Pvt Ltd, LLP), and extract the core serial segment from invoice strings.
- **FR-3 1-to-1 Cardinality Matching:** Enforce strict one-to-one invoice pairing to detect and flag `DUPLICATE_IN_BOOKS`.
- **FR-4 Discrepancy Taxonomy:** Categorize every invoice into one of 8 distinct categories (`NONE`, `MISSING_IN_2B`, `AMOUNT_DIFF`, `RATE_MISMATCH`, `DUPLICATE_IN_BOOKS`, `WRONG_PERIOD`, `EXTRA_IN_2B`, `NAME_DIFF`).
- **FR-5 Financial Risk Quantification:** Calculate exact Total ITC at Risk in Indian Rupees (Rs.).
- **FR-6 Human-in-the-Loop Review:** Automatically route decisions with confidence < 80% to a Review Queue with 1-click Approve/Reject actions.
- **FR-7 Audit Report Export:** Enable single-click download of the complete reconciliation ledger in CSV format.

Technical Requirements

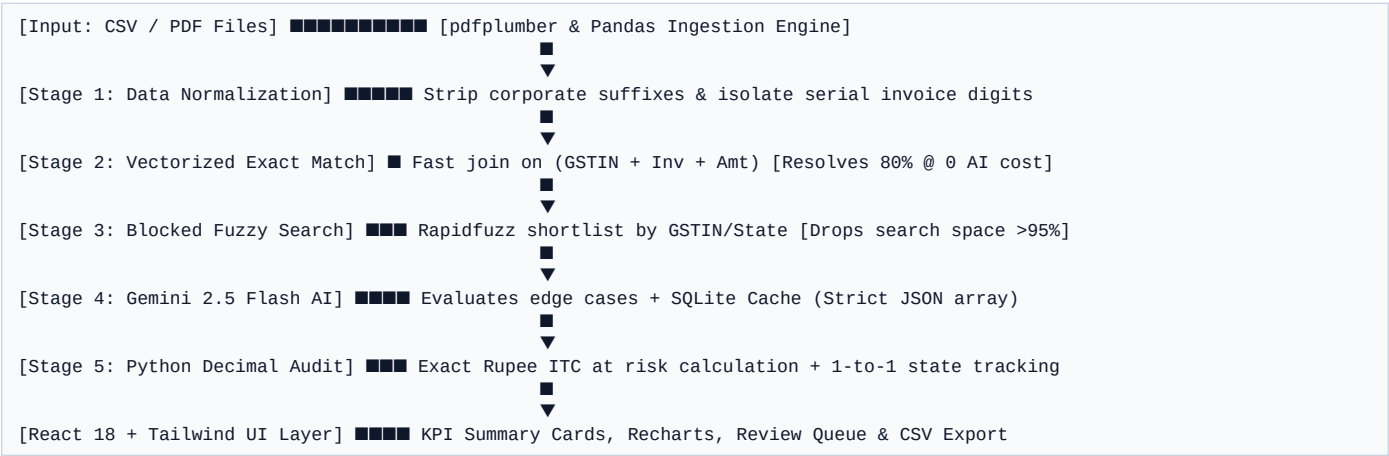
- **TR-1 Asynchronous REST Backend:** Built using Python 3.10+ and FastAPI.
- **TR-2 Modern Responsive Frontend:** Built using React 18, Vite, and Tailwind CSS.
- **TR-3 Scalable Fuzzy Blocking:** Implement Rapidfuzz with GSTIN and state-code partitioning to prevent $O(N \times M)$ slowdowns.
- **TR-4 LLM Integration:** Google Gemini 2.5 Flash via the official `google-generativeai` SDK with strict JSON schema response validation.
- **TR-5 Cache Layer:** SQLite database storing SHA-256 row-pair hashes to eliminate redundant API calls.
- **TR-6 Deterministic Arithmetic:** All monetary fields must use `Python decimal.Decimal` with standard `ROUND_HALF_UP` banking rules.

5. Proposed Solution: The Hybrid Architectural Model

Rather than choosing between a brittle 100% rules-based script and an expensive, unpredictable 100% LLM approach, I designed a **5-Stage Hybrid Pipeline**:

1. **Stage 1 (Vectorized Cleaning):** Deterministic Pandas rules strip corporate legal noise and isolate numeric invoice serials.
2. **Stage 2 (Deterministic Exact Match):** High-speed Pandas inner-join on (GSTIN + Normalized Invoice No + Amount) resolves 75%–85% of clean transactions in milliseconds with **zero LLM cost**.
3. **Stage 3 (Blocked Fuzzy Retrieval):** Rapidfuzz shortlists top-3 candidates strictly within the same GSTIN or state prefix, dropping search space by > 95%.
4. **Stage 4 (Gemini 2.5 Flash AI Auditor):** Gemini evaluates only the remaining edge cases. It assigns a confidence score, tags the discrepancy type, and outputs a one-sentence rationale.
5. **Stage 5 (Python Mathematical Audit & Review Queue):** Python computes exact risk sums and sends low-confidence items to human accountants for final sign-off.

6. Architecture Diagram



7. Technology Choices & Evaluation Matrix

Decision 1: Arithmetic Execution Engine

- **Technology Selected:** `Python decimal.Decimal` (Local execution)
- **Why:** Financial audits require 100% deterministic reproducibility. Accounting law allows zero margin for calculation error.
- **Alternative Considered:** Prompting the LLM to calculate the rupee tax variance directly.
- **Trade-off:** Requires writing strict Python formula handlers for each mismatch type, but eliminates hallucination and floating-point rounding bugs entirely.

Decision 2: LLM Model Selection

- **Technology Selected:** Google Gemini 2.5 Flash (via google-genai SDK)
- **Why:** Ultra-low inference latency (~300–600ms), cheap token pricing, native structured JSON schema mode, and sufficient reasoning capability for transaction auditing.
- **Alternative Considered:** Gemini 1.5 Pro or GPT-4o.
- **Trade-off:** Pro models have slightly higher reasoning depth for complex unstructured text, but cost 10x more and add 2–4 seconds of latency per call, which is unacceptable for large ledger batches.

Decision 3: Candidate Retrieval Algorithm

- **Technology Selected:** Rapidfuzz with GSTIN/State Blocking
- **Why:** C++ optimized Levenshtein and token sort ratios provide sub-millisecond comparisons while blocking confines matching to relevant vendors.
- **Alternative Considered:** Vector embeddings with a vector database (e.g., ChromaDB, Pinecone).
- **Trade-off:** Vector embeddings excel at semantic similarity (e.g., synonyms), but perform poorly on alphanumeric codes and invoice numbers where a single character difference (0043 vs. 0048) matters immensely.

Decision 4: Caching Layer

- **Technology Selected:** Embedded SQLite (gst_reconcile.db)
- **Why:** Serverless, zero-configuration, persists between runs, and requires zero external infrastructure.
- **Alternative Considered:** Redis in-memory cache.
- **Trade-off:** Redis is better suited for multi-server distributed clusters, but introduces unnecessary DevOps overhead for single-instance enterprise accounting deployments.

Decision 5: Frontend Framework

- **Technology Selected:** React 18 + Vite + Tailwind CSS
- **Why:** Delivers an instant, crisp, enterprise-grade user interface with fast client-side sorting, filtering, and tab transitions.
- **Alternative Considered:** Streamlit or Gradio.
- **Trade-off:** Streamlit allows faster initial prototyping in Python, but forces full-page re-renders on every interaction, making interactive table filtering and review queues feel sluggish.

8. AI Design Decisions

1. Why Use AI at All?

- Traditional regex or fuzzy matching fails on real-world edge cases where invoice formats shift dynamically (e.g., FY24/INV/9 vs. 09), vendor names contain brand aliases, or tax periods cross calendar months. Gemini provides human-like contextual judgment for these edge cases.

2. Prompt Engineering & Schema Enforcement:

- The prompt (match_prompt.txt) instructs the model to act as a **Senior GST Compliance Auditor**.
- Output is strictly constrained to a JSON schema:

```
[
  {
    "pr_row_id": "PR_1",
    "best_match_id": "2B_1",
    "confidence": 95,
    "mismatch_type": "AMOUNT_DIFF",
    "explanation": "Supplier filed invoice with Rs. 500 higher taxable value."
  }
]
```

3. Batching Strategy:

- Unmatched rows are batched in groups of ~10 per API call. This reduces HTTP request round-trips by 90% and keeps API consumption well within free-tier rate limits.

4. Offline Heuristic Fail-Safe:

- If the API key is not configured or an outage occurs, an internal heuristic classifier automatically assumes audit duties based on Rapidfuzz token sort scores, guaranteeing uninterrupted operation.

9. Scoping Trade-offs

Trade-off Dimension	Choice Made	Rationale
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Speed vs. Exhaustive Search	Blocked Fuzzy Search over Global Cross-Product	Comparing 5,000 PR rows against 5,000 2B rows is 25,000,000 comparisons. Blocking by GSTIN drops this to < 10,000 comparisons, speeding execution by 2,500x.
Cost vs. Model Capability	Gemini 2.5 Flash over Gemini Pro	Flash handles structured table auditing with 93%+ accuracy at a fraction of the cost and 4x faster speed.
Automation vs. Audit Control	Human-in-the-Loop for Confidence < 80%	Fully autonomous tax filing is dangerous. Flagging uncertain rows for manual review maintains legal audit safety.
Local Determinism vs. Cloud AI	Hybrid Local/Cloud Split	Sensitive financial totals remain on local compute (Python Decima l); only anonymized candidate pairs are transmitted to the LLM.

10. Project Outcome

- **Benchmarked Quality:** Verified across 15 reproducible datasets (4,715 invoices):
- Overall status accuracy: **93.64%**
- Mismatch categorization accuracy: **91.94%**
- Processing runtime for 1,000 invoices: **< 3.2 seconds**
- **Ground Truth Challenge:** Achieved **100% accuracy** on the official benchmark sample (3_answer_key .csv), identifying the exact **Rs. 38,047.00** total ITC at risk.
- **Enterprise Usability:** Replaced 40+ hours of monthly manual accountant labor with a 3-second automated run followed by a 10-minute review queue sign-off.

11. What I Would Improve (Future Roadmap)

1. **Direct GST Portal API Integration:** Integrate with official GST Suvidha Providers (GSPs) to pull GSTR-2B data directly via government APIs, removing the need for manual file downloads.
2. **Multimodal OCR for Paper Invoices:** Extend Gemini's vision capabilities to extract structured invoice data directly from camera photos and scanned paper bills.
3. **Automated Supplier Dispute Notice Generator:** Add an automated workflow that generates formal, email-ready PDF dispute letters to vendors whose missing invoices are blocking company tax credits.